

MRI reporting system

IMPROD (version 1.0)

In the IMPROD clinical trial, T2-weighted images (T2wi) and diffusion weighted imaging (DWI) data, acquired in three separate acquisitions, are being obtained.

The overall probability (= Likert score) of each lesion being a cancer is reported to the physician performing the targeted biopsy. Each lesion with a Likert score of 3 or more is marked in the simplified scheme and location described in the text. Representative images (T2wi, DWI) are attached to the report.

The Likert score is defined as follows:

- 1: Significant cancer is highly unlikely to be present
- 2: Significant cancer is unlikely to be present
- 3: Significant cancer is possible (equivocal finding)
- 4: Significant cancer is likely to be present
- 5: Significant cancer is highly likely to be present

The Likert score is based on the combined evaluation of T2wi, apparent diffusion coefficient (ADC) maps, trace DWI images acquired using b value of 1500 s/mm², and trace DWI images acquired using b value of 2000 s/mm².

The final Likert score is NOT based on an arithmetical combination of ratings for each method (T2wi, DWI), but on the overall probability of a lesion being malignant or not.

The Likert score is an “overall impression score” of a lesion to be a significant cancer or not.

Supplementary information

The following information is supplied to assist in the Likert score decision process for each lesion.

1A. T2-weighted images, based on PI-RADS version 1.0 reference [1] (below):

T2wi for the peripheral zone:

1. Uniform high signal intensity (SI)
2. Linear, wedge shaped, or geographic areas of lower SI, usually not well demarcated
3. Intermediate appearances not in categories 1/2 or 4/5
4. Discrete, homogeneous low signal focus/mass confined to the prostate
5. Discrete, homogeneous low signal intensity focus with extra-capsular extension/invasive behavior or mass effect on the capsule (bulging), or broad (>1.5 cm) contact with the surface

T2wi for the transition zone (TZ)

1. Heterogeneous TZ adenoma with well-defined margins: “organized chaos”
2. Areas of more homogeneous low SI, however well demarcated, originating from the TZ/BPH
3. Intermediate appearances not in categories 1/2 or 4/5
4. Areas of more homogeneous low SI, ill defined: “erased charcoal sign”
5. Same as 4, but involving the anterior fibromuscular stroma or the anterior horn of the PZ, usually lenticular or water-drop shaped.

1B. T2-weighted images, based on reference [2] (below):

T2wi for the peripheral zone:

1. Homogenous high SI.
2. Linear area perpendicular to the prostate capsule.
3. Area of lower SI not well demarcated.
4. Discrete homogenous low SI nodular lesion.
5. Discrete homogenous low SI nodular lesion with mass effect.

T2wi for the central gland:

1. Spherical lesion with well-defined margins and heterogeneous signal intensity
2. Spherical or non-spherical lesion with well-defined margins, heterogeneous SI.
3. Spherical lesion with well-defined margins, homogenous low SI
4. Lenticular or ill-defined lesion with low SI.
5. Lenticular or ill-defined lesion with low SI and mass effect.

2. Diffusion weighted imaging (DWI)

2A. Qualitative, as defined in PI-RADS version 1.0 reference [1] (below)

DWI for the peripheral and transition zones:

1. No reduction in ADC compared with normal glandular tissue. No increase in SI on any high b-value image ($\geq b800$)
2. Diffuse, hyper SI on $\geq b800$ image with low ADC; no focal features; however, linear, triangular or geographical features are allowed
3. Intermediate appearances not in categories 1/2 or 4/5
4. Focal area(s) of reduced ADC but iso-intense SI on high b-value images ($\geq b800$)
5. Focal area/mass of hyper SI on the high b-value images ($\geq b800$) with reduced ADC

2B. Quantitative, defined below, is applicable only for DWI protocol as used in the current trial - see details in reference [2] (below)

Multiple circular regions of interests (ROIs) with size of 21 mm^2 should be drawn on the ADC maps. If the mean ADC values in at least one ROI (without the presence of susceptibility artifacts) is below the cut-off value of $1250 \times 10^{-6} \text{ mm}^2/\text{s}$ in the peripheral zone and $1100 \times 10^{-6} \text{ mm}^2/\text{s}$ in the central gland, a lesion is considered to be a PCa lesion (score of 3 or more in the scoring system below).

The scoring system is based on the mean value calculated over a region of interest with a size of 21 mm^2 placed on a parametric (=voxel by voxel fit) ADC map (generated using a noise index of 20 using the Siemens Leonardo Workstation):

ADC for the peripheral zone:

1. above $1450 \times 10^{-6} \text{ mm}^2/\text{s}$
2. $1450 - 1350 \times 10^{-6} \text{ mm}^2/\text{s}$
3. $1350 - 1250 \times 10^{-6} \text{ mm}^2/\text{s}$
4. $1250 - 1150 \times 10^{-6} \text{ mm}^2/\text{s}$
5. below $1150 \times 10^{-6} \text{ mm}^2/\text{s}$

ADC for the central gland:

1. above $1300 \times 10^{-6} \text{ mm}^2/\text{s}$
2. $1300 - 1200 \times 10^{-6} \text{ mm}^2/\text{s}$
3. $1200 - 1100 \times 10^{-6} \text{ mm}^2/\text{s}$
4. $1100 - 1000 \times 10^{-6} \text{ mm}^2/\text{s}$

5. Below $1000 \times 10^{-6} \text{ mm}^2/\text{s}$

Prediction of a lesion's Gleason score

Prediction of a lesion's Gleason score is based on the ADC map and trace DWI images acquired using b values of 1500 s/mm^2 and 2000 s/mm^2 .

Dominant Gleason grade 4 (= Gleason score of 4+3 or higher) is:

1. probable:

- high signal intensity is present on trace DWI images acquired using b values of 1500 s/mm^2 and 2000 s/mm^2 and ADC values, performed according to reference [2] (below)

850 $\times 10^{-6} \text{ mm}^2/\text{s}$

2. highly probable:

- high signal intensity is present on trace DWI images acquired using b values of 1500 s/mm^2 and 2000 s/mm^2 and ADC values, performed according to reference [2] (below), below **750** $\times 10^{-6} \text{ mm}^2/\text{s}$

NOTES:

- carefully distinguish between restricted signal on trace DWI images acquired using b values of 1500 s/mm^2 and/or 2000 s/mm^2 from susceptibility artifacts caused by inhomogeneous B0 field (eg. rectal gas)

- benign prostatic hyperplasia (BPH) can have high signal intensity on b values of 1500 s/mm^2 and/or 2000 s/mm^2 and low ADC values. However, BPH typically has well-defined margins with a fibrotic capsule having a low signal, and heterogeneous signal intensity on T2wi.

References:

1. Barentsz JO, Richenberg J, Clements R, et al. ESUR prostate MR guidelines 2012. *Eur Radiol* 2012; **22**:746-57.
2. Jambor I, Kahkonen E, Taimen P, et al. Prebiopsy multiparametric 3T prostate MRI in patients with elevated PSA, normal digital rectal examination, and no previous biopsy. *J Magn Reson Imaging* 2015; **41**:1394-404.